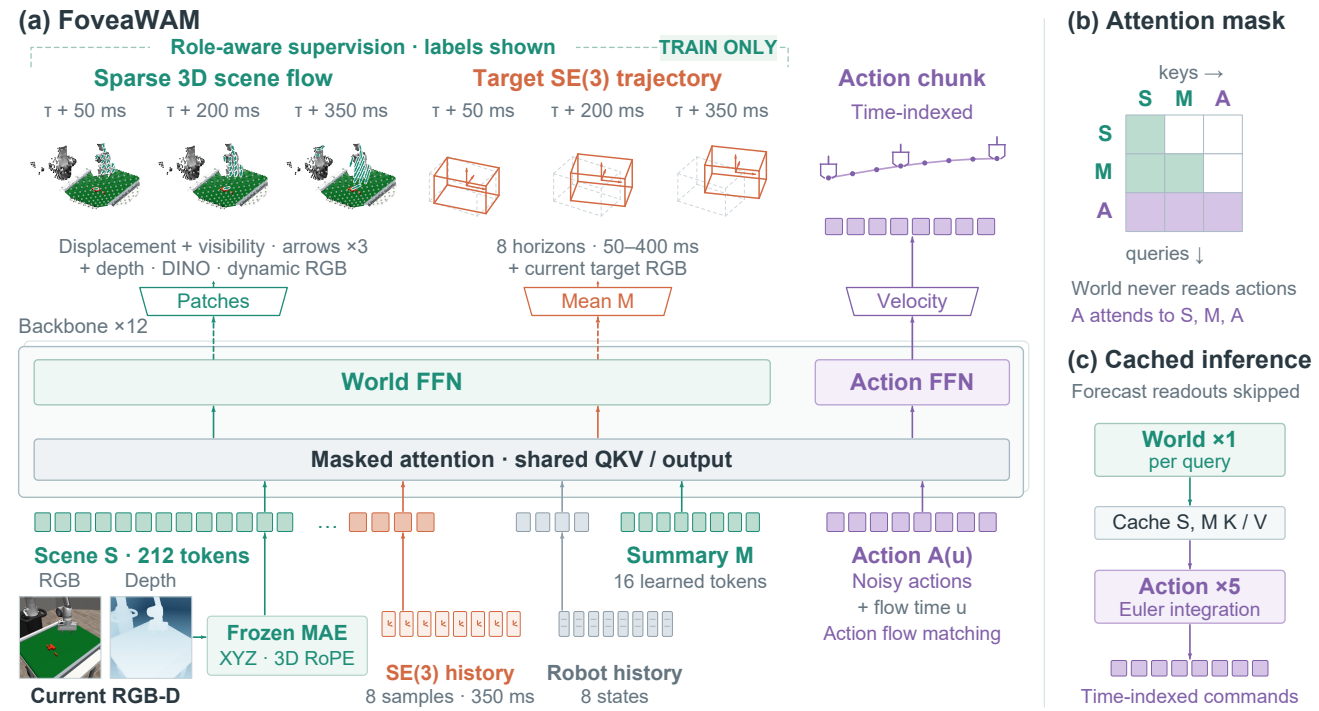

(a) FoveaWAM
Role-aware supervision · labels shown
TRAIN ONLY
Sparse 3D scene flow
τ + 50 ms
τ + 200 ms
τ + 350 ms
Target SE(3) trajectory
τ + 50 ms
τ + 200 ms
τ + 350 ms
Action chunk
Time-indexed
Displacement + visibility · arrows ×3
+ depth · DINO · dynamic RGB
8 horizons · 50–400 ms
+ current target RGB
Patches
Mean M
Velocity
Backbone ×12
World FFN
Action FFN
Masked attention · shared QKV / output
Scene S · 212 tokens
RGB
Depth
Summary M
16 learned tokens
Action A(u)
Noisy actions
+ flow time u
Action flow matching
Frozen MAE
XYZ · 3D RoPE
SE(3) history
8 samples · 350 ms
Robot history
8 states
Current RGB-D
(b) Attention mask
keys →
S
M
A
S
M
A
queries ↓
World never reads actions
A attends to S, M, A
(c) Cached inference
Forecast readouts skipped
World ×1
per query
Cache S, M K / V
Action ×5
Euler integration
Time-indexed commands